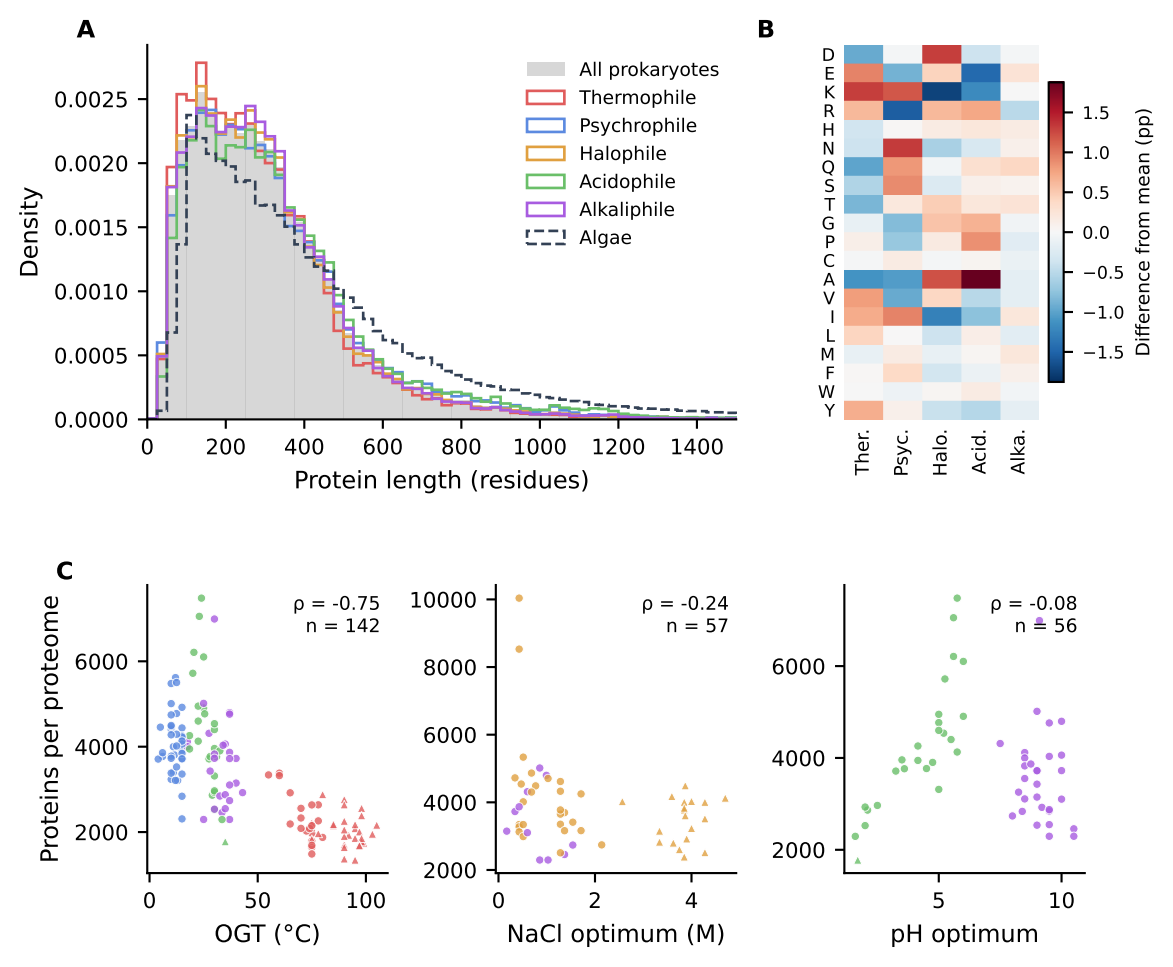

A
Density
0.0025
0.0020
0.0015
0.0010
0.0005
0.0000
0
200
400
600
800
1000
1200
1400
Protein length (residues)
All prokaryotes
Thermophile
Psychrophile
Halophile
Acidophile
Alkaliphile
Algae
B
D
E
K
R
H
N
Q
S
T
G
P
C
A
V
I
L
M
F
W
Y
Ther.
Psyc.
Halo.
Acid.
Alka.
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
Difference from mean (pp)
C
Proteins per proteome
6000
4000
2000
ρ = -0.75
n = 142
0
50
100
OGT (°C)
10000
8000
6000
4000
2000
ρ = -0.24
n = 57
0
2
4
NaCl optimum (M)
6000
4000
2000
ρ = -0.08
n = 56
5
10
pH optimum